

ePiE

exposure to Pharmaceuticals in the Environment

User Manual

Version 1.0

July 2020

© Radboud University Nijmegen, 2020

Parts of this end user manual may be reproduced, provided acknowledgement is given to Radboud University Nijmegen, along with the title and year of publication.

Contact:

Prof. dr. A.M.J. Ragas

a.ragas@fnwi.ru.nl

This is a publication of:

Radboud University Nijmegen

Department of Environmental Science

P.O. Box 9010 | 6500 GL Nijmegen

The Netherlands

Acknowledgement:

This user manual was developed by Jolinde Kettelarij, PhD and Cornelis van Loon, MSc (Caesar Consult) based on Deliverable 3.1.1 from the iPiE project and Oldenkamp et al. (2018), with input from Rik Oldenkamp, PhD and Professor Ad Ragas (Radboud University Nijmegen).

The research leading to the ePiE model has received support from the Innovative Medicines Initiative Joint Undertaking under iPiE grant agreement n° 115735, resources of which are composed of financial contribution from the European Union's Seventh Framework Programme (FP7/2007-2013) and EFPIA companies' in kind contribution.

Contents

List of abbreviations	3
1. Introduction.....	4
1.1. Context	4
1.2. ePiE application domain.....	4
2. Input parameters.....	9
2.1. Overview.....	9
2.2. Detailed parameter description	10
2.2.1. Substance identifiers	10
2.2.2. Physicochemical parameters.....	10
2.2.3. Metabolism parameters.....	11
2.2.4. Environmental partitioning parameters.....	11
2.2.5. Degradation process parameters.....	11
2.2.6. Consumption data	13
3. User interface	14
4. Output: interpretation of results	22
5. Training material	25
6. Further reading.....	27
7. References.....	28
Appendix 1 – Tables training material.....	30

List of abbreviations

API	Active Pharmaceutical Ingredient
ePiE	Exposure to Pharmaceuticals in the Environment (model)
iPiE	Intelligence-led Assessment of Pharmaceuticals in the Environment (project)
IMI	Innovative Medicines Initiative
HydroSHEDS	Hydrological data and maps based on SHuttle Elevation Derivatives at multiple Scales
QSAR	Quantitative Structure-Activity Relationship
p.e.	population equivalents
UWWTD	Urban Waste Water Treatment Directive
WWTP	Wastewater Treatment Plant
FLO1K	Streamflow dataset at 1 km resolution
DDD	Defined Daily Dosages

1. Introduction

1.1. Context

ePiE (exposure to Pharmaceuticals in the Environment) is a model developed to estimate environmental concentrations of active pharmaceutical ingredients (APIs) throughout Europe. It is a spatially explicit model that predicts API concentrations in surface waters (rivers and lakes) and their upper sediment layers at specific coordinates or at a specific distance downstream of the point of emission.

The model is integrated in iPiE-SYS (<https://ipiesys.eu/>); a system that was developed in the frame of IMI's iPiE (Innovative Medicines Initiative; Intelligence-led Assessment of Pharmaceuticals in the Environment) project. A detailed description of background and use of QSARs in the ePiE model can be found in the "ePiE Technical model description – version 1", which can be found through the website of the iPiE project (<http://i-pie.org/>).

1.2. ePiE application domain

A flow diagram of ePiE (Figure 1) shows the input flow of API-specific parameters, following the selection of an API and a river basin of interest.

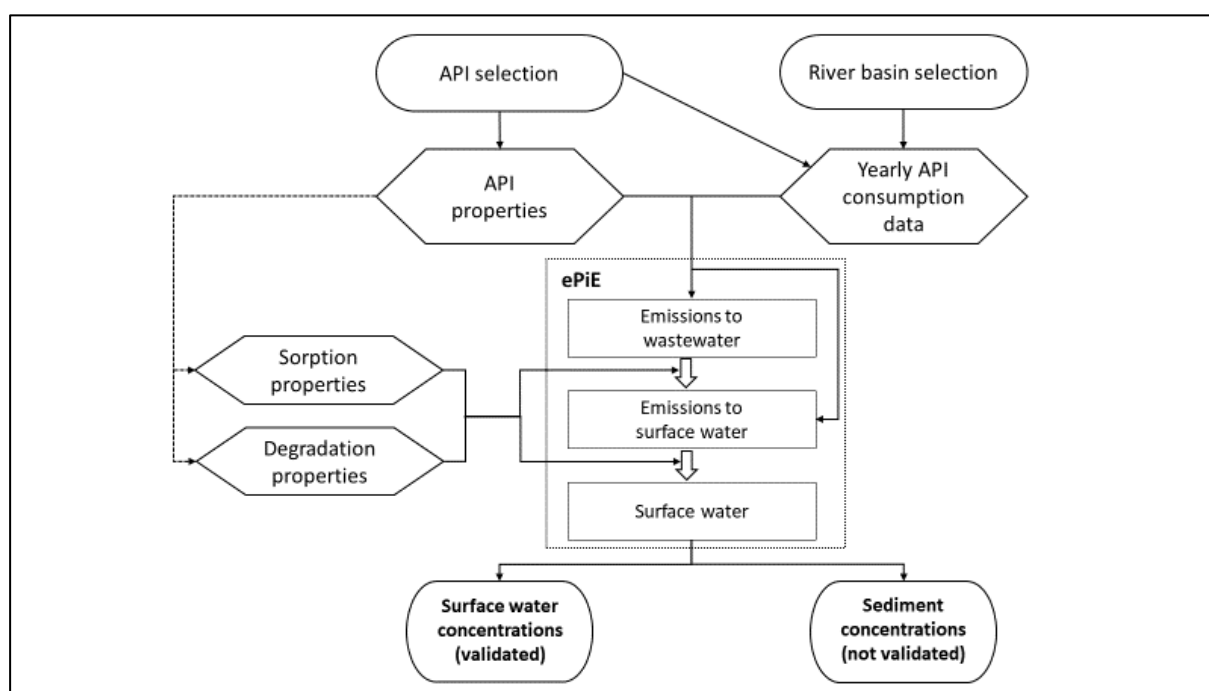


Figure 1. Flow diagram of ePiE within iPiE-SYS, showing the relationship between the input of several API-specific properties and calculations in ePiE, ending with estimated environmental API concentrations.

The ePiE model requires the input of API substance identifiers and physicochemical parameters. These are used to run QSARs that are integrated in ePiE, for example to estimate environmental partitioning parameters when these are not available.

Central to the ePiE model is a set of network nodes. River networks were defined as delineated by the global HydroSHEDS database (Lehner et al., 2008), and network nodes were allocated to river elements within a 30 arcseconds raster (~1 km). Via overlay of the river network with the lakes and reservoirs in the global database HydroLAKES (Messenger et al., 2016), intersections between the river network and lakes and reservoirs were determined, and were added as additional nodes to the

network. Based on the global HydroBASINS database(Lehner and Grill, 2013), which only includes basins below 60 N, the ePiE model defines which subset of network nodes belongs to a river basin of interest. River basins are pre-set in ePiE, and at the moment the model is restricted to the basins that were used in validation exercises:

- Ouse (York/Leeds) (UK)
- Wear (UK)
- Dove (UK)
- Rhine (Switzerland, Germany, the Netherlands, Austria, Belgium, France, Luxembourg)
- Odense (Denmark)
- Turia (Spain)
- Guadalquivir (Spain)
- Santiago (Spain)

In conjunction with selection of the river basin, ePiE requires the input of yearly national consumption data (kg/year) for the API of interest in the country or countries that the river basin of interest covers. ePiE does not allow for the input of regional consumption data, since administrative or political boundaries and hydrological boundaries and sewerage service areas often do not properly align. Furthermore, ePiE assumes equal per capita consumption, and cannot account for age or group differences.

To estimate the emission of APIs to wastewater, the consumption data needs to be adjusted by accounting for the fractions of the administered dose excreted as parent compound (and reversible conjugates) via urine and faeces. In addition, ePiE requires the fraction of any prodrug dosage that is excreted/egested as the API of interest (or as its reversible conjugate). At the moment, ePiE cannot estimate environmental concentrations of metabolites with the exception of the metabolites of prodrugs. In addition, dermally administered APIs, of which a part might enter wastewater directly after application, cannot specifically be estimated in ePiE. However, the user can incorporate knowledge on this fraction in the overall fraction excreted/egested unchanged. The user should also be aware that metabolism data can be highly variable.

Wastewater is usually indirectly emitted to surface water after passage through a wastewater treatment plant (WWTP), but might partly be discharged into surface water directly. In ePiE, yearly API loads in wastewater are distributed over agglomerations (i.e. urban human settlements comprising of built-up areas) in the relevant country, based on the UWWTD (Urban Waste Water Treatment Directive) database (EEA, 2015). Emissions are based on the generated wastewater loads of agglomerations (expressed in population equivalents; p.e.) as a proxy of their relative population size. The level of direct and indirect emission to surface water depends on the level of connectivity of agglomerations to WWTPs. ePiE has some restrictions related to the use of the UWWTD database. First, agglomerations with a generated load below 2,000 p.e. are not included in the database, since the UWWTD does not require reporting on them. This means that for areas with small agglomerations (<2,000 p.e.), environmental API concentrations may be underestimated by ePiE. Second, some WWTPs receive a combination of urban and industrial wastewater, of which the database only reports the combined load. API loads towards such WWTPs might therefore deviate from the estimations. Third, estimations may also deviate due to the fact that WWTP discharges are allocated to nearest water body, which may not always be the discharge location. Finally, due to missing data on agglomeration connectivity to WWTPs, ePiE predictions are less reliable for Southern Italy and Croatia.

When wastewater passes through a WWTP, the emission of APIs to surface water depends on the type of treatment techniques applied in the WWTP and the removal of APIs related to these techniques. If not provided by the UWWTD database, ePiE assumes a default WWTP design as

formulated in SimpleTreat 4.0 (Struijs, 2014), except for the size of the WWTP which is derived from the UWWTD database. The UWWTD database distinguishes between primary, secondary, and more advanced (tertiary) treatment techniques. Due to insufficient data availability on degradation rates and reactor kinetics for tertiary treatment techniques, the ePiE model currently cannot estimate removal of chemicals via these techniques. Therefore, emission to surface water could be overestimated by the ePiE model if the surface water is connected to a WWTP with tertiary treatment techniques.

After direct or indirect emission, ePiE predicts environmental concentrations of APIs in surface waters (rivers, lakes and reservoirs) and their upper sediment layers. In ePiE, the thickness of the upper sediment layer has a fixed value of 0.03 m. Each WWTP and each agglomeration with less than 100% WWTP connectivity was assigned a specific location where the emission to surface water occurs. APIs are followed through the river network using a routing procedure ordered from the most upstream to the most downstream nodes. As such, the contribution of all upstream emissions to local concentrations is considered.

The partitioning of APIs between the water column and a range of media (air and several types of solids) is incorporated in ePiE. Solids-water partitioning coefficients are estimated via the weighted summation of partitioning coefficients for the separate forms of the API (neutral and alternate, i.e. anionic or cationic). This is done in order to allow for location-specific parameterization due to geographical differences in aquatic pH. However, the potential for operationalization of a spatially explicit approach is currently still limited due to lacking spatially explicit data on environmental properties such as carbon content, temperature and pH. Several QSARs are incorporated in ePiE to estimate partitioning coefficients of neutral, anionic and cationic forms of the APIs. However, those QSARs have some limitations in their application domain, which are described in Table 1. For air-water partitioning, the coefficient is estimated regardless of ionization of the API. We expect this simplification to be of little influence, considering the relatively small role that volatilization has for these generally non-volatile compounds.

Several dissipation processes influence the estimated concentration of APIs in surface waters and their upper sediment layers. The ePiE model accounts for three degradation processes (biodegradation, photolysis and hydrolysis) and two advective processes (sedimentation and volatilization). In ePiE, several assumptions related to these dissipation processes are made that cannot be changed by the user. Biodegradation, photolysis and hydrolysis are all assumed to occur in surface waters. In sediments, photolysis is assumed negligible, and only biodegradation and hydrolysis are considered. Furthermore, ePiE only allows the input of an overall photolysis rate constant, due to lack of experimental data on indirect and direct photolysis separately. In addition, we conservatively assumed that degradation processes only relate to the available fraction of the API and that the fraction adsorbed to solids is not available for degradation.

In ePiE, estimated API concentrations can be predicted for different flow regimes, e.g. annual mean flow, and highest and lowest monthly mean flow. This is possible due to the implementation of the FLO1K dataset (Barbarossa et al., 2018) in ePiE. This dataset provides estimations of flow at a spatial resolution of 30 arc seconds (~1 km), which is consequently the highest resolution possible in ePiE. The user should be aware that flow regressions are uncertain under irregular conditions (e.g. abstractions).

The output of ePiE consists of API concentrations in surface water. At the moment, the ePiE model in iPiE-SYS cannot make estimations for terrestrial, groundwater or food chain concentrations. Prediction of sediment concentrations is included in the iPiE-SYS version of ePiE, but the accuracy of

the predicted sediment concentrations has not yet been checked based on a comparison with measurement data.

Table 1. Partitioning coefficients used in ePiE that are based on QSARs, and model restrictions related to these QSARs.

Parameter estimated by QSAR	QSAR(s) in Technical model description (v1)	QSAR application domain	Limitations in application domain	Reliability	Reference
$K_{OC,n}$	15	QSAR for neutral compounds. Based on data from 81 hydrophobic chemicals with $\log K_{ow}$ from 1 to 7.5. The equation had an r^2 of 0.887.	During validation, the smallest errors occurred for a $\log K_{ow}$ of 1 to 4.	The equation is considered reasonably reliable.	Sabljić et al., 1995
$K_{OC,n}$ and $K_{OC,alt}$	16 and 17	QSAR for anionizing compounds. Based on data from 93 acids with a pK_a between 0 and 12 and a $\log K_{ow,n}$ of -2 to 8. Equation 16 had an r^2 of 0.81 (n=11) and Equation 17 an r^2 of 0.04 (n=29). The combined equation had an r^2 of 0.54 for the training set (n=62) and an r^2 of 0.44 for the validation set (n=31).	Can introduce considerable uncertainty in estimating partitioning behaviour of acids, particularly strong acids.	QSAR performance is suboptimal. If available, empirical K_{oc} values measured at the relevant pH should be preferred.	Franco and Trapp, 2008
$K_{OC,n}$ and $K_{OC,alt}$	18 and 19	QSAR for cationic compounds. Based on data from 65 bases with a pK_a between 2 and 12 and a $\log K_{ow,n}$ of -2 to 7. Equation 38 had an r^2 of 0.83 (n=11) and Equation 37 an r^2 of 0.75 (n=13). The combined equation had an r^2 of 0.60 for the training set (n=43) and an r^2 of 0.44 for the validation set (n=22).	Can introduce considerable uncertainty in estimating partitioning behaviour of bases.	QSAR performance is suboptimal. If available, empirical K_{oc} values measured at the relevant pH should be preferred.	Franco and Trapp, 2008
Kp_{DOC}	20	QSAR for neutral compounds. Based on 127 data points from 14 nonionizable chemicals.	Can introduce considerable uncertainty in estimating partitioning behaviour.	If available, empirical Kp_{DOC} values should be preferred.	Burkhard, 2000
$K_{OW,app}$	21	QSAR for ionizing compounds. The equation was based on a very limited data set of carboxylic acid and 4 other anions.	Equation is based on the assumption that the K_{ow} of the ionized form is approximately a factor of 3.5 lower than that of the neutral form.	Empirical basis is very weak and empirical data should always be preferred over this QSAR estimate.	Trapp and Horobin, 2005

Table 2. Fixed values used in the ePiE model to estimate environmental API concentrations

Parameter	Short description	Fixed value	Unit	Ref.
H_{sed}	Thickness of the upper sediment layer	0.03	m	Expert assessment
θ_{sed}	Porosity of sediment	0.8	L_{water}/L_{sed}	Paterson and Mackay, 1995
ρ_w	Density of water	1	kg_{water}/L_{water}	
$\rho_{s,sed}$	Mineral density of sediment solids	2.1633	kg_{solids}/L_{solids}	Fantke et al., 2016; Honti et al., 2016
C_{susp}	Concentration suspended solids in the water column	$0.015 \cdot 10^{-3}$	kg_{solids}/L_{water}	Humbert et al., 2011
C_{DOC}	Concentration dissolved organic carbon in the water column	$0.005 \cdot 10^{-3}$	kg_{DOC}/L_{water}	Asselman, 1997
T_{field}	Local temperature at which the degradation rate constant is applied	285; or georeferenced water temperatures	K	ECHA, 2016
D_λ	Ratio between the average path length of light with wavelength λ , and the average water depth	1.2 (non-turbid waters)	-	Schwarzenbach et al., 1993
$V_{sed,acc}$	Net sediment accumulation rate in water	$8.6 \cdot 10^{-11}$	m/s	Fantke et al., 2016
$V_{m,w,w sed}$	Partial mass transfer coefficient at the water side of the water/sediment interface	$2.778 \cdot 10^{-6}$	m/s	Mackay, 2001
$V_{m,sed,w sed}$	Partial mass transfer coefficient at the sediment side of the water/sediment interface	$2.778 \cdot 10^{-8}$	m/s	Mackay, 2001
$V_{sed,susp}$	Settling velocity of suspended particles	$2.89 \cdot 10^{-5}$	m/s	den Hollander et al., 2004
$f_{V_{solid,sd}}$	Volume fraction solids in sediment	0.2	-	Paterson and Mackay, 1995

2. Input parameters

2.1. Overview

A number of input parameters can be entered in iPiE-SYS to configure the ePiE model (Table 3), of which some are required (* in Table 3) to run the model.

Table 3. Input parameters in the ePiE model used to estimate environmental API concentrations

Name	Short description
<i>Substance identifiers</i>	
API*	Name of Active Pharmaceutical Ingredient
API Prodrug	Name of prodrug
CAS*	CAS number
<i>Physicochemical parameters</i>	
Class*	Chemical class of the API
MW*	Molecular weight
P _v *	Vapour pressure at 25 °C
S*	Solubility in water at 25 °C
K _{OW} *	Octanol/water partitioning coefficient of the neutral form
pK _a *	Acid dissociation coefficient
<i>Metabolism parameters</i>	
f _u *	Fraction of dose excreted unchanged via urine
f _f *	Fraction of dose egested unchanged via faeces
f _{Prodrug}	Fraction of the administered prodrug egested as the API of interest or as its reversible conjugates via urine/faeces
<i>Environmental partitioning parameters</i>	
K _{p_ps}	Sewage-solids/water partitioning coefficient
K _{p_as}	Activated sludge-solids/water partitioning coefficient
K _{p_susp}	Suspended matter-solids/water partitioning coefficient
K _{p_DOC}	Dissolved organic carbon/water partitioning coefficient
K _{p_sd}	Sediment-solids/water partitioning coefficient
K _{oc}	Organic carbon/water partitioning coefficient
<i>Degradation process parameters</i>	
k _{bio,wwtp}	Pseudo-first order biodegradation rate constant secondary treatment
k _{bio,sw1}	Pseudo-first order experimental biodegradation rate constant in surface water
T _{bio,sw}	Temperature at which k _{bio,sw1} was derived
k _{bio,sd1}	Pseudo-first order experimental biodegradation rate constant in sediment
T _{bio,sd}	Temperature at which k _{bio,sd1} was derived
k _{hydro,sw}	Pseudo-first order experimental hydrolysis rate constant in surface water
T _{hydro,sw}	Temperature at which k _{hydro,sw} was derived
k _{hydro,sd}	Pseudo-first order experimental hydrolysis rate constant in sediment
T _{hydro,sd}	Temperature at which k _{hydro,sd} was derived
λ _{solar}	Maximum absorption wavelength for solar radiation
k _{photo12,sw}	Pseudo-first order experimental photolysis rate constant in surface water
T _{photo12,sw}	Temperature at which k _{photo12,sw} was derived
<i>Consumption data</i>	
River basin*	Pre-set river basins in ePiE. At the moment this is restricted to the basins that were used in validation exercises (see Section 1.2)
Yearly API consumption*	Yearly consumption of API (or prodrug) of interest per country, based on the selected river basin

*Parameter input required to run the ePiE model

For some parameters, default values exist or estimations are calculated using QSAR models or other models integrated in iPiE-SYS. In some cases, parameter values are available from the iPiE database

that is integrated in iPiE-SYS. In Chapter 2.2, each group of parameters is explained in more detail, including default values.

2.2. Detailed parameter description

The ePiE model follows the exposure assessment chain, combining information at different levels of spatial resolution with physicochemical properties of the API assessed. The input parameters that are used to run the model are described in detail below. For some APIs, a distinction can be made between the **neutral** and **alternate** (ionized) forms of the API. If so, the user needs to enter two separate values for some parameters (e.g. partitioning and degradation); one for the neutral form and one for the alternate form. ePiE cannot simulate APIs that exist in more than two forms; in this case, the user is advised to run a simulation for the neutral and dominant alternate form.

2.2.1. Substance identifiers

Name	Short description	Default value	Unit
API	Name of Active Pharmaceutical Ingredient	-	-
API Prodrug	Name of prodrug	-	-
CAS	CAS number	-	-

As a starting point, the ePiE model requires the input of substance identifiers. These are not used in calculations, but only used for identification of the API. When the API of interest (**API**) is formed as a metabolite from another API, that is, its prodrug, the name and other data for that prodrug are also needed. The prodrug must be entered as a separate API (**API Prodrug**). The CAS number (**CAS**) of the API is needed as additional identification parameter.

2.2.2. Physicochemical parameters

Name	Short description	Default value	Unit
Class	Chemical class of the API	Neutral	-
MW	Molecular weight	-	g/mol
P _v	Vapour pressure at 25 °C	1.00E-10	Pa
S	Solubility in water at 25 °C	1.00E+03	mg/L
K _{ow}	Octanol/water partitioning coefficient of the neutral form	-	-
pK _a	Acid dissociation coefficient	-	-

The chemical **class** and the acid dissociation coefficient (**pK_a**) of the API are used to estimate fractions of the neutral and alternate forms of the API. For chemical class, the choice can be made between neutral, acid or base. Other alternate forms (e.g. zwitterionic) cannot be modelled with ePiE. For substances having multiple pK_as it is suggested to use the pK_a value that is closest to the pH value of the local environment (i.e., current default 7.4 for sediments and surface water and 7.0 for primary and secondary treatment).

The molecular weight (**MW**), vapour pressure (**P_v**), and the solubility in water (**S**) of the API are used to calculate the volatilization of the API from the water column to the air. For APIs that are salts, the molecular weight of the free acid or base should be entered. MW and pK_a can be derived from the chemical structure of the molecule. The same is true for the octanol/water partitioning coefficient (**K_{ow}**) of the neutral form of the API. This parameter is used to predict partitioning behaviour and the degradation rate of the API.

2.2.3. Metabolism parameters

Name	Short description	Default value	Unit
f_u	Fraction of dose excreted unchanged via urine	100%	-
f_f	Fraction of dose egested unchanged via faeces	0%	-
$f_{Prodrug}$	Fraction of the administered prodrug egested as the API of interest or as its reversible conjugates via urine/faeces	100%	-

API loads entering the wastewater are estimated using information on human metabolism. After oral consumption, APIs can be absorbed to different extents from the intestinal tract, entering the blood circulation. The non-absorbed fraction is egested with the faeces and ends up in the wastewater. The absorbed fraction might partly be metabolized in the body, and partly be excreted via urine, or via bile into the faeces. To estimate the load entering the wastewater, the consumption thus needs to be adjusted by accounting for the fraction of the administered dose excreted as parent compound (and reversible conjugates) via urine (f_u), and the fraction of the administered dose excreted/egested as parent compound (and reversible conjugates) via faeces (f_f). The sum of these fractions is used in ePiE to calculate the API load reaching wastewater. In addition, ePiE requires the fraction of any prodrug dosage that is excreted/egested as the API of interest (or as its reversible conjugate) ($f_{Prodrug}$).

2.2.4. Environmental partitioning parameters

Name	Short description	Default value	Unit
K_{p_ps}	Sewage-solids water partitioning coefficient	-	L_{water}/kg_{solids}
K_{p_as}	Activated sludge-solids water partitioning coefficient	-	L_{water}/kg_{solids}
K_{p_susp}	Suspended matter-solids water partitioning coefficient	-	L_{water}/kg_{solids}
K_{p_DOC}	Dissolved organic carbon water partitioning coefficient	-	L_{water}/kg_{solids}
K_{p_sd}	Sediment-solids water partitioning coefficient	-	L_{water}/kg_{solids}
K_{OC}	Organic carbon water partitioning coefficient	-	L_{water}/kg_{solids}

The partitioning of APIs between a range of media and water is described at different points throughout the ePiE model. ePiE requires the input of solids-water partitioning (K_p ; L/kg) values for five different types of solids-water matrices: primary sewage (K_{p_ps}), secondary sludge (K_{p_as}), suspended matter (K_{p_susp}), dissolved organic carbon (K_{p_DOC}), and sediments (K_{p_sd}).

If the user wants to make a distinction between the **neutral** and **alternate** form of the API, two separate values are needed for each of the above partitioning coefficients; one for the neutral form and one for the alternate form. During model calculations these are then combined to sorption of the total compound at local pH. If the API only exists in a neutral form, or if the user does not want to make such a distinction, only one value per parameter is needed.

When no direct K_p values are provided as input (either for the total compound or for the separate neutral and ionized forms), the values are currently estimated under the assumption that sorption is solely driven by organic carbon content of the medium assessed. Consequently, partitioning coefficients are automatically estimated by ePiE, based on the organic-carbon/water partitioning coefficient (K_{OC}) or the octanol/water partitioning coefficient (K_{OW}). In case of APIs that are ionizing, a total K_{OC} is based on the K_{OC} s of the individual forms ($K_{OC,n}$, $K_{OC,alt}$; kg/L).

2.2.5. Degradation process parameters

Name	Short description	Default value	Unit
$k_{bio,wwtp}$	First order biodegradation rate constant secondary treatment	0	s^{-1}
$k_{bio,sw1}$	Pseudo-first order experimental biodegradation rate constant in surface water	0	s^{-1}
$T_{bio,sw}$	Temperature at which $k_{bio,sw1}$ was derived	293.15	K

$k_{bio,sd1}$	Pseudo-first order experimental biodegradation rate constant in sediment	0	s^{-1}
$T_{bio,sd}$	Temperature at which $k_{bio,sd1}$ was derived	293.15	K
$k_{hydro,sw}$	First order experimental hydrolysis rate constant in surface water	0	s^{-1}
$T_{hydro,sw}$	Temperature at which $k_{hydro,sw}$ was derived	293.15	K
$k_{hydro,sd}$	First order experimental hydrolysis rate constant in sediment	0	s^{-1}
$T_{hydro,sd}$	Temperature at which $k_{hydro,sd}$ was derived	293.15	K
λ_{solar}	Maximum absorption wavelength for solar radiation	298*	nm
$k_{photo12,sw}$	Pseudo-first order experimental photolysis rate constant in surface water	0	s^{-1}
$T_{photo12,sw}$	Temperature at which $k_{photo12,sw}$ was derived	293.15	K

* At 298 nm the attenuation coefficient is highest, i.e. 0.0430 cm^{-1} . The highest attenuation coefficient results in the lowest light penetration in water, hence a conservative estimate of photolysis.

The ePiE model accounts for five separate processes that result in dissipation of APIs from environmental media, being the water column and the upper sediment layer. Three of these five dissipation processes are degradation processes, i.e. **biodegradation**, **photolysis** and **hydrolysis**. Two of the five dissipation processes are advective (transport) processes, i.e. **sedimentation** (water to sediment) and **volatilization** (water to air). The two advective dissipation processes are estimated automatically by ePiE, based on several physicochemical parameters (described in paragraph 3.2.2 of the Technical Manual) and fixed default values. The three degradation processes are estimated by ePiE using a combination of user input and fixed default values. Due to fixed default values, the application domain of the ePiE model has some restrictions, as described in the Introduction.

To account for the degradation processes, ePiE requires the user input of several parameters. API removal during secondary (activated sludge) treatment is calculated using the input of a pseudo first-order biodegradation rate constant ($k_{bio,wwtp}$).

Within the ePiE model, biodegradation, photolysis and hydrolysis are all assumed to occur in surface waters. In sediments, photolysis is assumed negligible, and only biodegradation and hydrolysis are considered. Pseudo-first order experimental biodegradation rate constants $k_{bio,sw1}$ and $k_{bio,sd1}$ are used to calculate biodegradation in surface water and the upper sediment layer, respectively.

Without input of biodegradation rate constants, no biodegradation will be assumed.

Similarly, the pseudo-first order experimental hydrolysis rate constant in surface water ($k_{hydro,sw}$) and in sediment ($k_{hydro,sd}$) are used to estimate hydrolysis of the API in surface water and sediment, respectively. Without input of these hydrolysis rate constants, no hydrolysis will be assumed. To correct the biodegradation and hydrolysis rates in surface water and sediment to local temperatures, ePiE requires the input of the temperature at which the rates were derived ($T_{bio,sw}$, $T_{bio,sd}$, $T_{hydro,sw}$ and $T_{hydro,sd}$).

To estimate photolysis that is adapted to local conditions in surface water, ePiE requires input of a pseudo-first order experimental photolysis rate constant in surface water ($k_{photo12,sw}$), the maximum absorption wavelength for solar radiation (λ_{solar}) and the temperature at which rate constant was derived ($T_{photo12,sw}$). Without input of the photolysis rate constant, no photolysis will be assumed.

Again, if the user wants to distinguish between the **neutral** and **alternate** form of the API, two separate values are needed for each of the parameters in this section; one for the neutral form and one for the alternate form.

2.2.6. Consumption data

Name	Short description	Default value	Unit
River basin*	Pre-set river basins in ePiE. At the moment this is restricted to the basins that were used in validation exercises.	Ouse, Wear, Dove, Rhine, Odense, Turia, Guadalquivir, Santiago	-
Yearly API consumption*	Yearly consumption of API (or prodrug) of interest per country, based on the selected river basin	1	kg/year

The model requires input of yearly consumption data (kg/year) in the country or countries that the river basin of interest covers. If data are only available in another form (i.e. the number of Defined Daily Dosages (DDD) or packages sold), the user needs to translate these into the total national amounts based on demographics. For APIs that are formed as an active metabolite of another API, i.e., their prodrug, consumption data for that prodrug are also required.

3. User interface

The ePiE model has been implemented in the iPiE-SYS platform: a web-based software application for end users, integrating the major results from the iPiE project, i.e. a database with thousands of properties relevant for environmental risk assessment for hundreds of active pharmaceutical ingredients (APIs), several tools for the predicting the environmental fate of APIs and several tools for predicting ecotoxicological effects. The description of the interface that follows below focuses on the ePiE model and does not include the other functionalities included in iPiE-SYS. A complete description of the iPiE-SYS platform can be found in iPiE deliverables 6.3 and 6.4 (Marusczyk et al. 2019). Part of the text that follows is based on this publication.

3.1 Access to iPiE-SYS

iPiE-SYS is a web application available online at <https://ipiesys.eu>. The user accesses the application through a standard web browser (such as Internet Explorer, Google Chrome or Mozilla Firefox) which connects to the iPiE-SYS server. Access is currently limited to partners of the iPiE consortium. Access to the system requires a username and a password, which is provided by the maintainer of the system (partner MN-AM) on a per-user basis.

3.2 Start page

Once logged in, the start page appears (Figure 1). The top row of the application is available at any stage or page of iPiE-SYS. The iPiE-SYS logo on the left side in the top row can be used to go back directly to the start page from any other page of the application.

On the left part of the top row the following links are provided.

- iPiE Project to open the iPiE project web site in a new browser tab;
- Terms & Privacy to review the terms of use and privacy to which the user agreed on when signing up for the system;
- Account name to manage the account including login of the system, changing the password or deleting the account (similar to iPiE Sum, see <https://ipiesum.eu/>);
- Help to display some help pages on the prediction models available in iPiE-SYS.

On the start page, the following functionality is available.

- Quick Search
- Full Search and Prediction Models
- Data Manager

The 2nd option (Full Search & Prediction Models) has to be selected to go to the ePiE model.

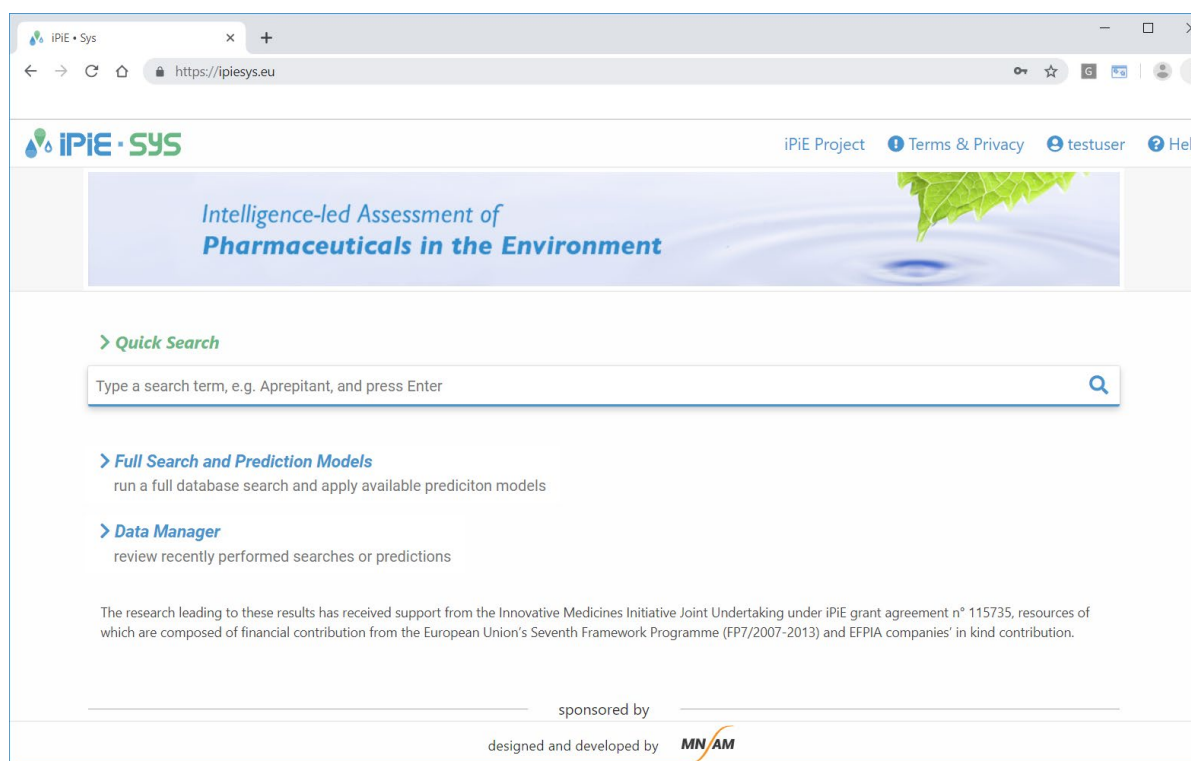


Figure 2: Start page of iPiE-SYS

3.3 Access to the ePiE model

The ePiE model is available in the iPiE-SYS platform **after** the end user has performed a database search for the API under investigation. There are several ways to search for APIs. Here, we will perform a text search for Carbamazepine as example (Figure 3a and 3b).

- Click “Text Search” in the Chemistry box (left panel);
- Click “partial”;
- Enter the search string “carbamazepine”;
- Click the “Apply” button in the lower right corner (Figure 3a);
- Now, the chemical structure of the compound appears. Select this structure and click the “Apply” button in the lower right corner again (Figure 3b).

The results of this search are shown in Figure 4.

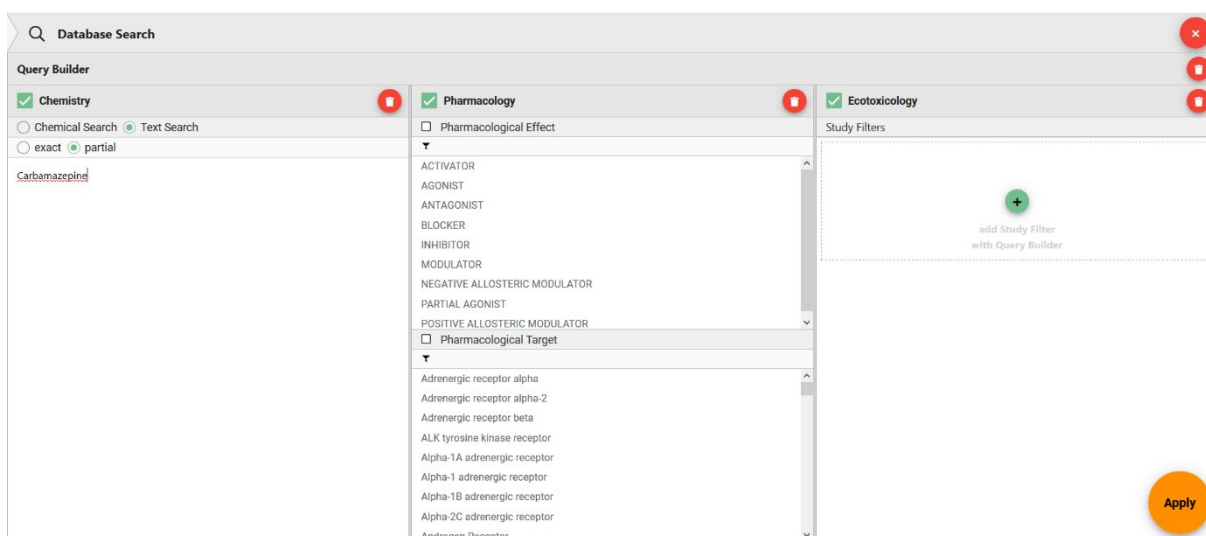


Figure 3a: Database search for carbamazepine (search input).

Figure 3b: Database search for carbamazepine (search output).

Figure 4: Result of the database search for carbamazepine.

3.4 Configuring the ePiE model

The user can now select “Configure ePiE Model” from the Actions panel (right panel), resulting in an empty configuration panel to be opened at the right side of the iPiE-SYS window (Figure 5).

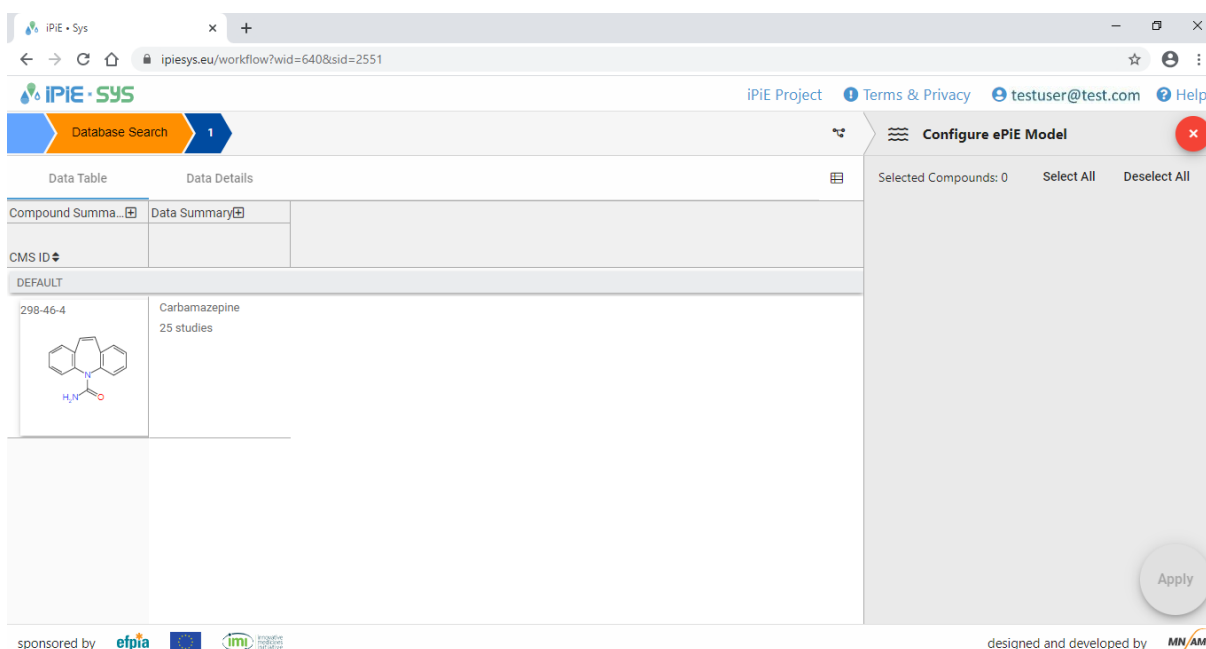


Figure 5: Empty configuration panel of the ePiE model.

Before the ePiE model can be configured, the user must first click on the molecular structure of the selected API – in this case carbamazepine – and then on the orange “Apply” button in the right lower

corner (Figure 6). A new field is now added to the main window with the header “ePiE model – Input Parameters” where the user can specify physicochemical parameters, select a river basin and add consumption data (Figure 7).

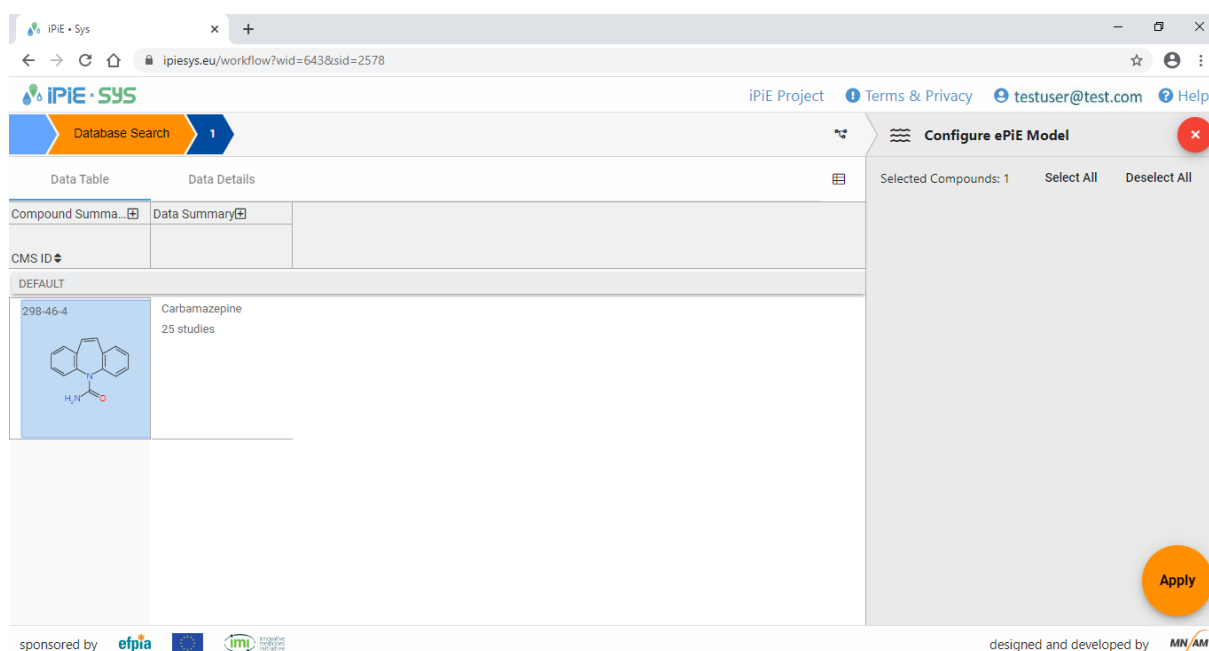


Figure 6: Before ePiE can be configured, one has to select the molecular structure of the API (in this case carbamazepine) and click the orange “Apply” button in the right lower corner.

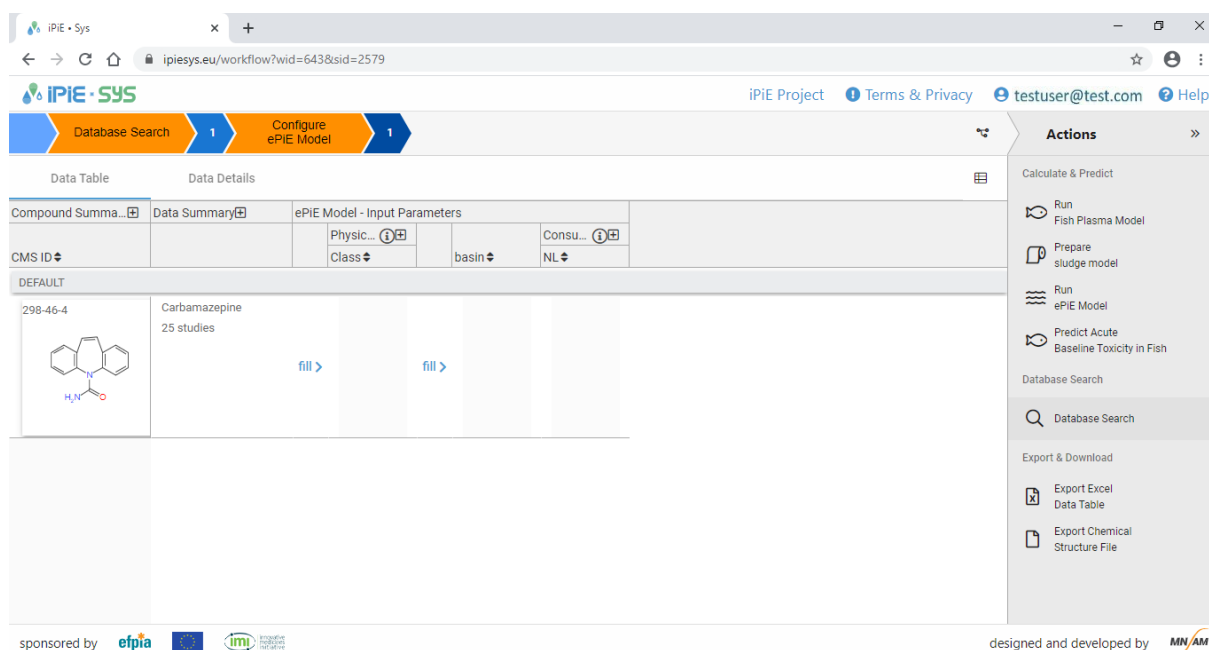


Figure 7: After clicking the Apply button, a new field labelled “ePiE Model – Input Parameters” is added to the main window.

When clicking on the first “Fill” button (on the left under Physicochemical Parameters), a new window opens where the user can enter input values for physicochemical parameters (Figure 8).

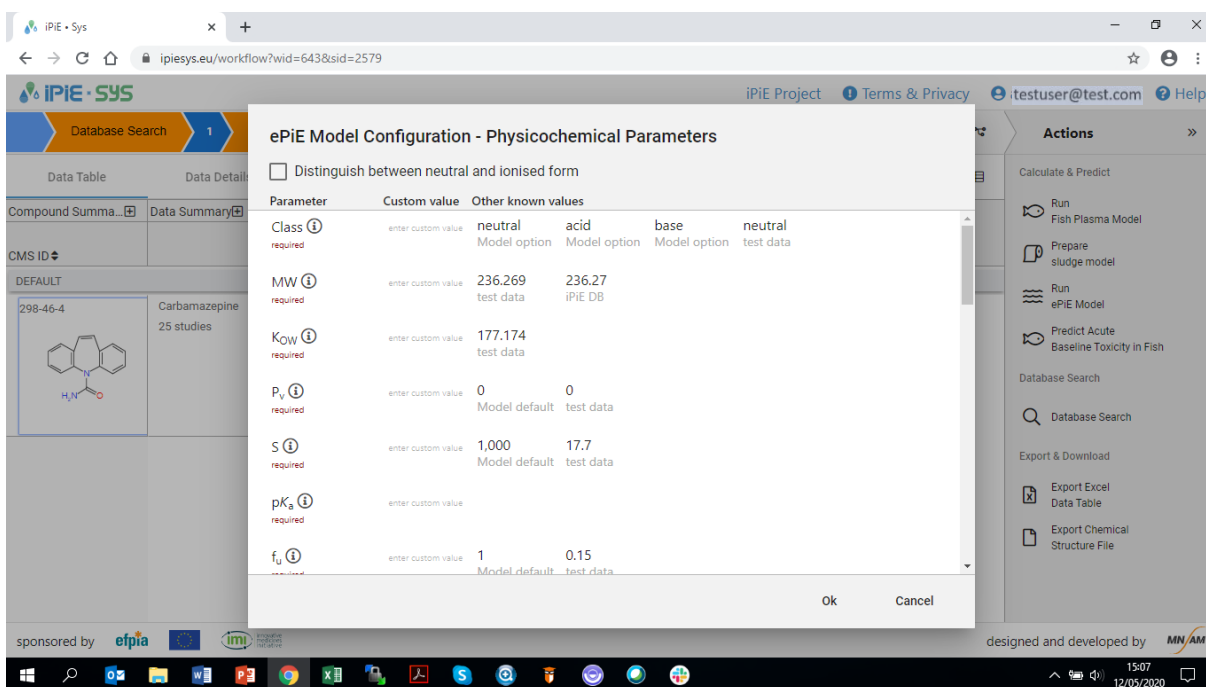


Figure 8: Pop-up window for entering physicochemical parameters for running the ePiE model.

The first choice the user has to make is whether or not the model should distinguish between the neutral and alternate (i.e., ionised) form of the API by ticking the box before “Distinguish between neutral and ionised form”. If this block is ticked, two values have to be entered for each parameter prone to ionisation, e.g. partition coefficients and degradation rates, i.e. one for the neutral form and one for the alternate, ionised form.

The window for entering physicochemical parameters has the form of a table, with a row for each input parameter. The first column of the table specifies the symbol of the parameter and indicates whether an input value is “required” (if not, it is optional). This column also contains a clickable ⁱ symbol for additional information on the parameter. This will open the Model Help function in the background (i.e., the right panel of the **main window**; Figure 9). The Model Help function provides a short description of each parameter, what it is used for in the ePiE model, its units and its default value (if applicable).

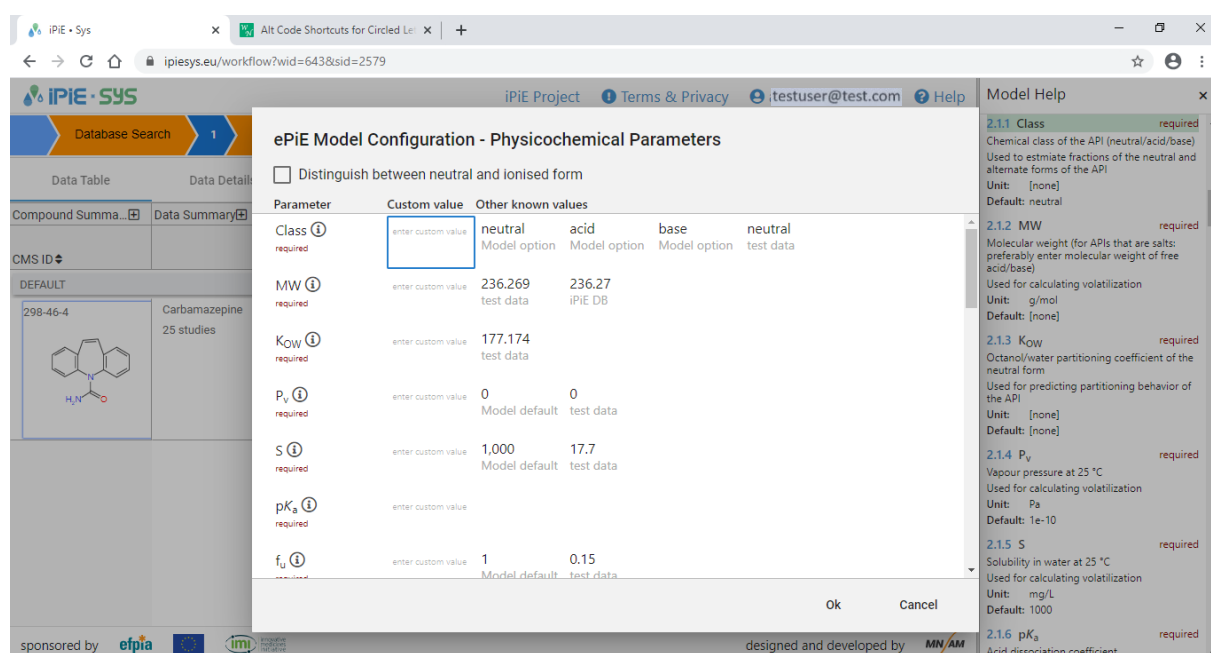


Figure 9: When clicking on the ① symbol of an input parameter (in this case the Class parameter), the Model Help window opens in the background (right panel of main window) providing a short description of each parameter, what it is used for in the ePiE model, its units and its default value (if applicable).

The second column of the input table allows the user to enter a custom value for each input parameter. Subsequent columns (labeled as “Other known values”) of the input table list any other available values, e.g. predefined model options, default values, values stored in the iPiE database or data extracted from literature by the model developers (“test data”).

To select and/or enter an input value, the user must first click on the relevant cell in the input table. If the cell for custom values is selected, the user must manually enter an input value. In all other cases, the predefined value in the selected cell will be used. Figure 10 shows an example with selected values for carbamazepine.

Note: The pKa is marked as a required parameter, but it is not. You can leave this cell empty for non-ionisable substances and ignore the subsequent error message.

Note: Do not enter “0” (zero) if data on pK_a, K_{ow} and other partition coefficients are lacking, but leave the cell empty. If you enter zero values, this will result in erroneous calculations. For degradation rates, you can enter a zero value reflecting the assumption of no degradation.

Warning: For the API Prodrug parameter, do not select the cell “None – Model default”. This will result in undefined results. This is a bug in the current version of iPiE-SYS.

ePiE Model Configuration - Physicochemical Parameters

☐ Distinguish between neutral and ionised form

Parameter	Custom value	Other known values			
Class [ⓘ] required	enter custom value	neutral Model option	acid Model option	base Model option	neutral test data
MW [ⓘ] required	enter custom value	236.269 test data	236.27 iPiE DB		
K _{OW} [ⓘ] required	enter custom value	177.174 test data			
P _v [ⓘ] required	enter custom value	0 Model default	0 test data		
S [ⓘ] required	enter custom value	1,000 Model default	17.7 test data		
pK _a [ⓘ] required	enter custom value				
f _u [ⓘ] required	enter custom value	1 Model default	0.15 test data		
f _r [ⓘ] required	enter custom value	0 Model default			
K _{p,ps} [ⓘ]	enter custom value	116.02 test data			
K _{p,as} [ⓘ]	enter custom value	588.6 test data			
K _{p,susp} [ⓘ]	enter custom value				
K _{p,DOC} [ⓘ]	enter custom value				
K _{p,sd} [ⓘ]	enter custom value	12.4 test data			
K _{OC} [ⓘ]	enter custom value				
k _{bio,wwtp} [ⓘ]	enter custom value	0 Model default	0 test data		
k _{bio,sw1} [ⓘ]	enter custom value	0 Model default	0 test data		
T _{bio,sw} [ⓘ]	enter custom value	293.15 Model default	293.15 test data		
k _{bio,sd1} [ⓘ]	enter custom value	0 Model default			
T _{bio,sd} [ⓘ]	enter custom value	293.15 Model default			
k _{hydro,sw} [ⓘ]	enter custom value	0 Model default	0 test data		
T _{hydro,sw} [ⓘ]	enter custom value	293.15 Model default	293.15 test data		
k _{hydro,sd} [ⓘ]	enter custom value	0 Model default			
T _{hydro,sd} [ⓘ]	enter custom value	293.15 Model default			
k _{photo12,sw} [ⓘ]	enter custom value	0 Model default	0 test data		
T _{photo12,sw} [ⓘ]	enter custom value	293.15 Model default	293.15 test data		
λ _{solar} [ⓘ]	enter custom value	298 Model default	285 test data		
API Prodrug [ⓘ]	enter custom value	None Model default			
f _{Prodrug} [ⓘ]	enter custom value	1 Model default			

Ok Cancel

Figure 10: Selected physicochemical input values for carbamazepine.

Once all relevant physicochemical input values have been selected, the user can click “Ok”.

When clicking on the second “Fill” button (under basin/Consumption data), a new window opens where the user can select a river basin and provide consumption data for each country that the river runs through. Currently, the user can select one of the following river basins from the top menu in this window: Odense (Denmark), Wear (UK), Ouse (UK), Dove (UK), Rhine (NL, DE, CH, AT, BE, FR, LU), Santiago (ES), Turia (ES) and Guadalquivir (ES).

Once a basin is selected, the countries that the river runs through are automatically highlighted in the table below; national consumption data (in kg/year) must be provided for these countries. Again, the user has the option to enter a custom value or select one of the “other know values” collected by the model developers. Figure 11 shows the input window for simulating carbamazepine concentrations in the Odense basin (Denmark) based on a custom national consumption value of 1749 kg/year. The user should click “Ok” to save the values.

Parameter	Custom value	Other known values
BE	enter custom value	5,775.927 2011, 5,800.818 2012, 5,783.936 2013, 5,765.633 2014
FR	enter custom value	34,959.36 2011, 35,119.828 2012, 35,293.804 2013, 35,477.759 2014
DK	1749	
LU	enter custom value	268.744 2011, 274.883 2012, 278.884 2013, 283.454 2014

Figure 11: Selected input values for simulating carbamazepine concentrations in the Odense basin (Denmark) based on a custom national consumption value of 1749 kg/year.

3.5 Running the ePiE model

Once the ePiE model is configured, you can run it by selecting “Run ePiE Model” from the list of actions in the right panel. You then have to click on the molecular structure of the API (in this case, carbamazepine), and click the orange “Apply” button in the right lower corner.

In case of missing data (e.g. the pK_a), an dialog box may appear (Figure 12). Click “OK” to continue. The ePiE model will now run (Figure 13). This may take about 2 to 10 minutes, depending on the size of the river basin selected.

The screenshot shows the iPiE-SYS interface with the 'Run ePiE Model' button highlighted. A dialog box is open with the message: 'Compound 1: missing required property pKa'. The dialog box has an 'OK' button.

Figure 12. The dialog box that appears in case of missing data (e.g. the pK_a). Click “OK” to continue.

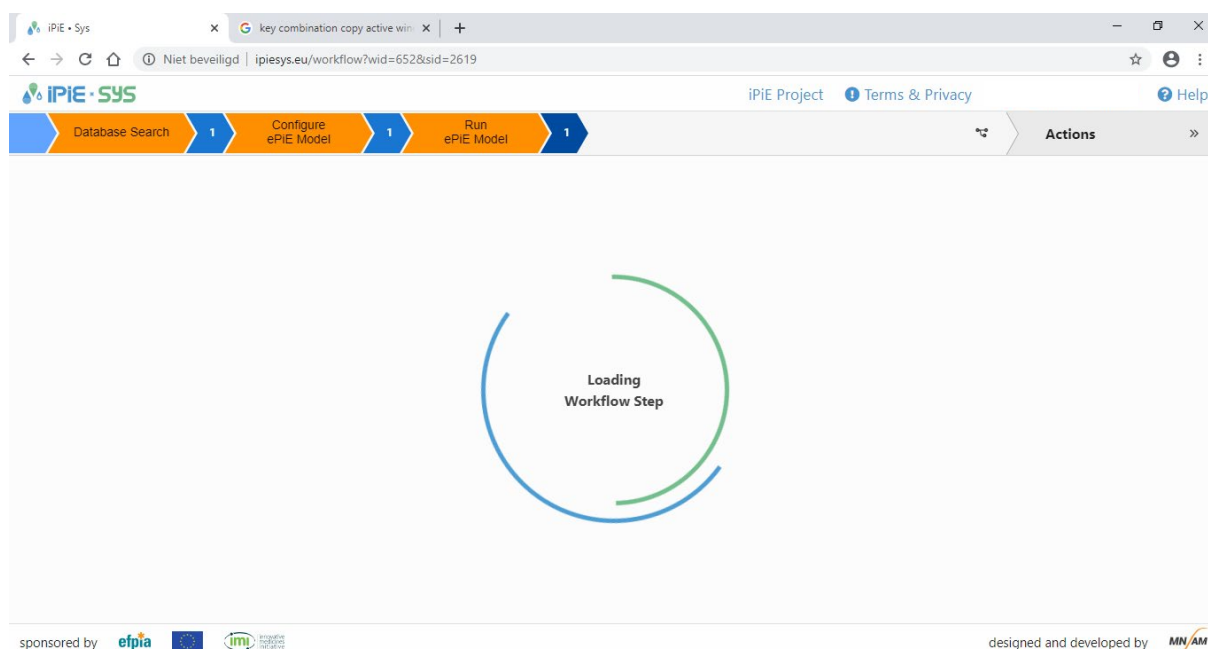


Figure 13: Screen while the ePiE Model is running.

4. Output: interpretation of results

Once ePiE is ready with its calculations, the original window reappears, but this now has an extra tab called “ePiE Map” under the blue/orange workflow bar (Figure 14).

Compound Summary	Data Summary	ePiE Map	Data Details	ePiE Model - Input Parameters
CMS ID 298-46-4 	Carbamazepine 25 studies	neutral fill >	Odense fill >	Success

Figure 14: Screen when ePiE is finished with its calculations.

To visualize the results, the user must click the “ePiE Map” tab and then on the molecule structure of the API (in this case, carbamazepine). An interactive map of Europe appears (Figure 15).

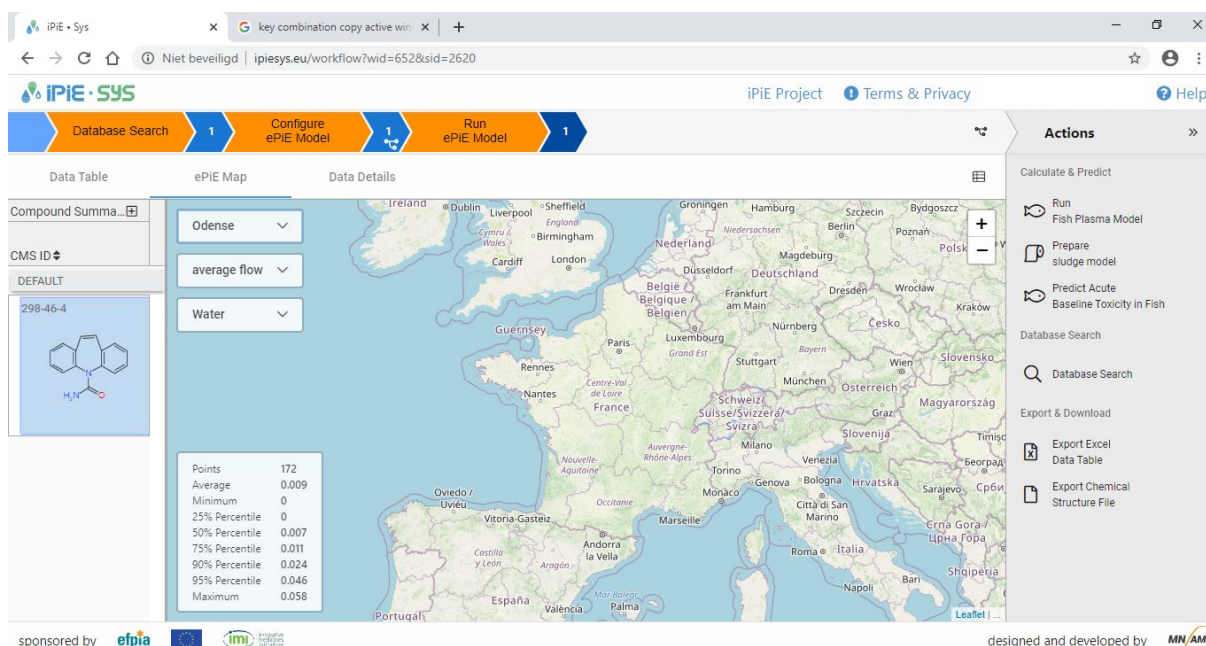


Figure 15: Interactive map of Europe.

On the left side of this map, the user can select:

- The river basin (if ePiE was run for multiple basins);
- The flow conditions of interest, i.e. average, minimum or maximum flow:
 - o The *minimum flow* indicates the lowest monthly average flow in 10 years for a basin,
 - o The *average flow* indicates the 10-year average flow for a basin,
 - o The *maximum flow* indicates the highest monthly average flow in 10 years for a basin.
- The compartment of interest, i.e. (surface) water or sediment.

In the left lower corner of the map, the number of nodes in the river basin is specified and some summary statistics on the predicted concentrations, i.e. the average, minimum, 25th, 50th, 75th, 90th and 95th percentiles, and the maximum.

The map of Europe can be zoomed and panned in order to focus on the area of interest. In our example, this is the Odense basin in Denmark. If we zoom in sufficiently, the nodes in the river basin appear as coloured dots on the map. Explanations for the colours can be found in Table 4. When the user hovers with the mouse over the dots on the map, the predicted concentration in the node appears (Figure 16).

Table 4. Type of node colours and explanations.

Colour of dot (node)	Explanation
Blue/Purple/Red	Dot indicating a predicted effect concentration (PEC) for a node in a basin: relatively low concentrations are blue, medium concentrations are purple, and relatively high concentrations are red
Black/Gray	No (zero) or undefined result

The colours *black* or *gray* for dots indicate errors in the calculation, such as missing data. Please check all input data first (see § 3.4) and run the iPiE Model again (see § 3.5).

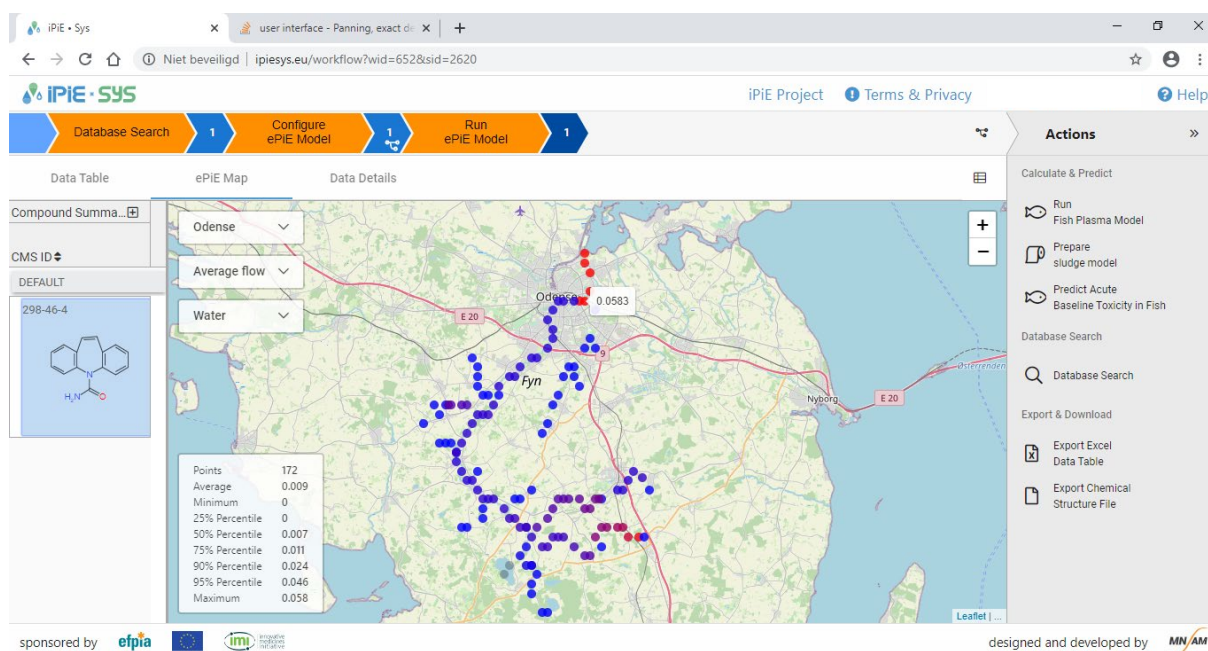


Figure 16: Map with predicted water concentrations of carbamazepine in the Odense river basin.

5. Training material

Case:

To prioritize your monitoring efforts, you are interested to predict environmental concentrations of **acetaminophen** (paracetamol), **desvenlafaxine**, **gabapentin**, **hydrocodone**, **loratadine** and **metformin** in the river **Ouse** basin (United Kingdom). The necessary input data for these APIs can be found in Appendix 1.

Exercise 1

- Add each API:
 - Press “Full Search and Prediction Models” on the home page of iPiE-SYS
 - Search the APIs by name or CAS number (Table A1). Once you have found the first API, find the next API by clicking “Database Search” in the “Actions” panel on the right.
- For each API, fill in the necessary physicochemical properties, metabolism parameters, environmental partitioning parameters, and degradation process parameters in the software (Table A1-A2-A3). In iPiE-SYS, default values or values from the iPiE database are available for some parameters.
- Select the river basin of interest and fill in the necessary consumption data (Table A4);
- Run the model to receive output for these APIs;
- Look at output (HTML-map).

Questions:

- What kind of information can you find on the interactive maps (surface water concentrations and sediment concentrations)?
- Where are the highest concentrations found? Does this make sense?
- In what order would you prioritize the monitoring of these APIs if you only consider their concentrations (and not their effects)?
- Now assume all consumption data for these APIs of interest are equal (10,000 kg/year). For which APIs the highest concentrations are predicted in water and/or sediment? Can you explain this based on the substance properties?

Exercise 2

Desvenlafaxine is not administered as a separate medication in the UK, but is only formed as a metabolite of its prodrug venlafaxine. Codeine functions as a prodrug for its metabolite hydrocodone.

- Add these two prodrugs as separate APIs:
 - Search each prodrug by name or CAS number (Table A5) by clicking “Database Search” in the “Actions” panel on the right.
- For each prodrug, fill in the necessary physicochemical properties, metabolism parameters, environmental partitioning parameters, and degradation process parameters in the software (Table A5-A6-A7). In iPiE-SYS, default values or values from the iPiE database are available for some parameters.
- Select the river basin of interest for the two prodrugs and fill in the necessary consumption data (Table A8);
- In the physicochemical parameters panel for hydrocodone fill in *Codeine* for “API Prodrug”, and *0,11* for f_{Prodrug} .
- In the physicochemical parameters panel for desvenlafaxine fill in *Venlafaxine* for “API Prodrug”, and *0,55* for f_{Prodrug} .

- Run the ePiE model to receive output for the two prodrugs and their metabolites.

Questions:

- Compare the output of the prodrugs to the output of the metabolites. What has changed for hydrocodone and desvenlafaxine when you added information about their prodrug?
- In what order would you now prioritize the monitoring of these APIs? (When based on predicted exposure concentrations (PEC) in surface waters: top 3 is metformin, gabapentin, acetaminophen. If toxic potency/predicted risks to fish is incorporated: top 3 is desvenlafaxine, loratadine, hydrocodone.)
- What happens when you change the river flow to high/low? Does that change the prioritization of the APIs? How?

Exercise 3: Compare your data to MEC data (Burns et al. 2018). What do you see? (Estimation of concentrations in smaller river stretches is less accurate).

6. Further reading

ePiE technical model description: <http://i-pie.org/epie/>

Scientific publication and supplementary information describing the ePiE model and validation of surface water concentration estimations: [Oldenkamp et al. 2018](#)

“iPiE_deliverable_D6.3_D6.4_predictiveModelIntegration_iPiEsysTool_v1_2019-07-11”, which can be found through <http://i-pie.org/epie/>.

7. References

- Asselman, N.E.M., 1997. Suspended sediment in the river Rhine: the impact of climate change on erosion, transport, and deposition, 234. Koninklijk Nederlands Aardrijkskundig Genootschap/Faculteit Ruimtelijke Wetenschappen Universiteit Utrecht.
- Barbarossa, V. et al., 2018. FLO1K, global maps of mean, maximum and minimum annual streamflow at 1 km resolution from 1960 through 2015. *Scientific Data*, 5: 180052.
- Burkhard, L.P., 2000. Estimating Dissolved Organic Carbon Partition Coefficients for Nonionic Organic Chemicals. *Environmental Science & Technology*, 34(22): 4663-4668.
- den Hollander, H.A., van Eijkeren, J.C.H., van de Meent, D., 2004. SimpleBox 3.0: Multimedia mass balance model for evaluating the fate of chemicals in the environment, National Institute for Public Health and the Environment (RIVM), Bilthoven, the Netherlands.
- ECHA, 2016. Guidance on information requirements and Chemical Safety Assessment. Chapter R.16: Environmental exposure assessment, European Chemicals Agency ECHA, Helsinki, Finland.
- EEA, 2015. UWWTD-WaterBase. Available from: <http://www.eea.europa.eu/data-and-maps/data/waterbase-uwwt-d-urban-waste-water-treatment-directive-4>. Latest update at 19 February 2015. European Environmental Agency.
- Fantke, P. et al., 2016. USEtox (R) 2.0 Documentation (Version 1), <http://usetox.org>. USEtox (R) is a registered trademark of the USEtox (R) Team in the European Union and the United States. All rights reserved. (C) USEtox (R) Team.
- Franco, A., Trapp, S., 2008. Estimation of the soil–water partition coefficient normalized to organic carbon for ionizable organic chemicals. *Environmental Toxicology and Chemistry*, 27(10): 1995-2004.
- Honti, M. et al., 2016. Bridging across OECD 308 and 309 Data in Search of a Robust Biotransformation Indicator. *Environmental Science & Technology*, 50(13): 6865-6872.
- Humbert, S. et al., 2011. Intake Fraction for Particulate Matter: Recommendations for Life Cycle Impact Assessment. *Environmental Science & Technology*, 45(11): 4808-4816.
- Lehner, B., Grill, G., 2013. Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. *Hydrological Processes*, 27(15): 2171-2186.
- Lehner, B., Verdin, K., Jarvis, A., 2008. New global hydrography derived from spaceborne elevation data. *Eos, Transactions American Geophysical Union*, 89(10): 93-94.
- Mackay, D., 2001. Multimedia Environmental Models: The Fugacity Approach. Toxicology and Environmental Health Series. CRC Press, Chelsea, Michigan, USA, 272 pp.
- Messenger, M.L., Lehner, B., Grill, G., Nedeva, I., Schmitt, O., 2016. Estimating the volume and age of water stored in global lakes using a geo-statistical approach. 7: 13603.
- National Health Service, 2017. Prescription Cost Analysis Data [online], available at: <<https://www.nhsbsa.nhs.uk/prescription-data/dispensing-data/prescription-cost-analysis-pca-data>>, last accessed: 11 Dec., 2017.
- Office for National Statistics, 2017. Population Estimates for UK, England and Wales, Scotland and Northern Ireland: Mid-2016. Available at: <https://www.ons.gov.uk/releases/populationestimatesforukenglandandwalesscotlandandnorthernirelandmid2016>, Fareham, UK.
- Paterson, S., Mackay, D., 1995. Interpreting chemical partitioning in soil-plant-air systems with a fugacity model. In: Trapp, S., McFarlane, J.C. (Eds.), *Plant Contamination*. Lewis Publishers, Boca Raton, Florida, USA, pp. 191-214.
- Sabljić, A., Güsten, H., Verhaar, H., Hermens, J., 1995. QSAR modelling of soil sorption. Improvements and systematics of log KOC vs. log KOW correlations. *Chemosphere*, 31(11–12): 4489-4514.
- Schwarzenbach, R.P., Gschwend, P.M., Imboden, D.M., 1993. Photochemical transformation reactions, *Environmental organic chemistry*. Wiley-Interscience, New York, New York, USA, pp. 436-484.

- Struijs, J., 2014. SimpleTreat 4.0: a model to predict fate and emission of chemicals in wastewater treatment plants, National Institute for Public Health and the Environment, Bilthoven, the Netherlands.
- Trapp, S., Horobin, R.W., 2005. A predictive model for the selective accumulation of chemicals in tumor cells. *European Biophysics Journal*, 34(7): 959-966.

Appendix 1 – Tables training material

Exercise 1

Table A1. Substance identifiers and physicochemical parameters for APIs used in this exercise.

API	API Prodrug	CAS	Class	MW (g/mol)	K _{ow,n}	P _v (Pa)	S (mg/L)	pK _a
Acetaminophen	-	103-90-2	acid	151.16	1.86	$2.59 \cdot 10^{-4}$	$1.40 \cdot 10^4$	9.46
Desvenlafaxine	Venlafaxine	93413-62-8	base	263.38	$5.25 \cdot 10^2$	$9.13 \cdot 10^{-6}$	$3.67 \cdot 10^3$	8.87
Gabapentin	-	60142-96-3	acid	171.24	$4.27 \cdot 10^{-2}$	$3.92 \cdot 10^{-8}$	$4.49 \cdot 10^3$	4.63
Hydrocodone	Codeine	125-29-1	base	299.36	$1.46 \cdot 10^2$	$1.61 \cdot 10^{-5}$	$1.79 \cdot 10^3$	8.61
Loratadine	-	79794-75-5	neutral	382.88	$4.54 \cdot 10^5$	$1.21 \cdot 10^{-6}$	$4.10 \cdot 10^3$	0
Metformin	-	657-24-9	base	129.16	$2.29 \cdot 10^{-3}$	$1.01 \cdot 10^{-2}$	$1.00 \cdot 10^6$	12.4

Table A2. Metabolism parameters and environmental partitioning parameters for APIs used in this exercise.

API	f _u	f _t	f _{Prodrug}	K _{p,ps} (L/kg)	K _{p,as} (L/kg)	K _{p,susp} (L/kg)	K _{p,DOC} (L/kg)	K _{p,sd} (L/kg)	K _{oc} (L/kg)
Acetaminophen	0.52	0	0	$5.11 \cdot 10^1$	$3.15 \cdot 10^2$	NA	NA	3.27	NA
Desvenlafaxine	0.65	0	0.55	$1.40 \cdot 10^2$	$1.01 \cdot 10^2$	NA	NA	NA	NA
Gabapentin	1.00	0	0	NA	NA	NA	NA	NA	NA
Hydrocodone	0.12	0	0.11	NA	$1.19 \cdot 10^2$	NA	NA	NA	NA
Loratadine	0.20	0	0	$2.34 \cdot 10^3$	$3.32 \cdot 10^3$	NA	NA	NA	NA
Metformin	1.00	0	0	NA	$1.62 \cdot 10^1$	NA	NA	NA	NA

Table A3. Degradation process parameters for APIs used in this exercise.

API	K _{bio,wwtp} (s ⁻¹)	K _{bio,sw1} (s ⁻¹)	T _{bio,sw} (K)	K _{bio,sd1} (s ⁻¹)	T _{bio,sd} (K)	k _{hydro,sw} (s ⁻¹)	T _{hydro,sw} (K)	k _{hydro,sd} (s ⁻¹)	T _{hydro,sd} (K)	k _{photo12,sw} (s ⁻¹)	T _{photo12,sw} (K)	λ _{solar} (nm)
Acetaminophen	$5.01 \cdot 10^{-4}$	$8.36 \cdot 10^{-6}$	293.15	0	293.15	0.00	293.15	0	293.15	$4.55 \cdot 10^{-6}$	293.15	250
Desvenlafaxine	0	$1.57 \cdot 10^{-8}$	293.15	0	293.15	$4.06 \cdot 10^{-7}$	293.15	0	293.15	$8.96 \cdot 10^{-6}$	293.15	275

Gabapentin	$1.15 \cdot 10^{-6}$	$8.02 \cdot 10^{-8}$	293.15	0	293.15	$1.33 \cdot 10^{-5}$	293.15	0	293.15	$2.00 \cdot 10^{-5}$	293.15	276
Hydrocodone	0	$4.34 \cdot 10^{-7}$	293.15	0	293.15	0	293.15	0	293.15	0	293.15	0
Loratadine	0	0	293.15	0	293.15	0	293.15	0	293.15	0	293.15	0
Metformin	$1.05 \cdot 10^{-4}$	$8.02 \cdot 10^{-8}$	293.15	0	293.15	$1.50 \cdot 10^{-5}$	293.15	0	293.15	$3.33 \cdot 10^{-7}$	293.15	232

Table A4. Yearly consumption data per API in England in 2016 (National Health Service, 2017) extrapolated to whole of United Kingdom based on demographics from Office for National Statistics (Office for National Statistics, 2017).

API	Consumption data (kg/year)
Acetaminophen	$2.52 \cdot 10^6$
Desvenlafaxine	0.00
Gabapentin	$1.60 \cdot 10^5$
Hydrocodone	$9.42 \cdot 10^3$
Loratadine	$8.58 \cdot 10^2$
Metformin	$1.15 \cdot 10^6$

Exercise 2

Table A5. Substance identifiers and physicochemical parameters for APIs used in this exercise.

API	API_metab	CAS	Class	MW (g/mol)	K _{ow,n}	P _v (Pa)	S (mg/L)	pK _a
Codeine	-	76-57-3	base	299.36	1.89*10 ¹	2.55*10 ⁻⁸	9.00*10 ³	9.19
Venlafaxine	-	93413-69-5	base	277.40	1.91*10 ³	3.28*10 ⁻⁵	5.72*10 ⁵	8.91

Table A6. Metabolism parameters and environmental partitioning parameters for APIs used in this exercise.

API	f _u	f _f	f _{Prodrug}	K _{p,ps} (L/kg)	K _{p,as} (L/kg)	K _{p,susp} (L/kg)	K _{p,DOC} (L/kg)	K _{p,sd} (L/kg)	K _{oc} (L/kg)
Codeine	0.40	0	0	NA	1.40*10 ¹	NA	NA	8.10	NA
Venlafaxine	0.05	0	0	1.50*10 ³	2.42*10 ²	NA	NA	NA	NA

Table A7. Degradation process parameters for APIs used in this exercise.

API	K _{bio,wwtp} (s ⁻¹)	K _{bio,sw1} (s ⁻¹)	T _{bio,sw} (K)	k _{bio,sd1} (s ⁻¹)	T _{bio,sd} (K)	k _{hydro,sw} (s ⁻¹)	T _{hydro,sw} (K)	k _{hydro,sd} (s ⁻¹)	T _{hydro,sd} (K)	k _{photo12,sw} (s ⁻¹)	T _{photo1,sw} (K)	λ _{solar} (nm)
Codeine	2.28*10 ⁻⁵	2.38*10 ⁻⁷	293.15	0	293.15	0	293.15	0	293.15	6.83*10 ⁻⁵	293.15	300
Venlafaxine	8.09*10 ⁻⁸	1.94*10 ⁻⁸	293.15	0	293.15	2.10*10 ⁻⁷	293.15	0	293.15	3.41*10 ⁻⁶	293.15	275

Table A8. Yearly consumption data per API in England in 2016(National Health Service, 2017) extrapolated to whole of United Kingdom based on demographics from Office for National Statistics(Office for National Statistics, 2017).

API	Consumption data (kg/year)
Codeine	5.62*10 ⁴
Venlafaxine	1.56*10 ⁴

